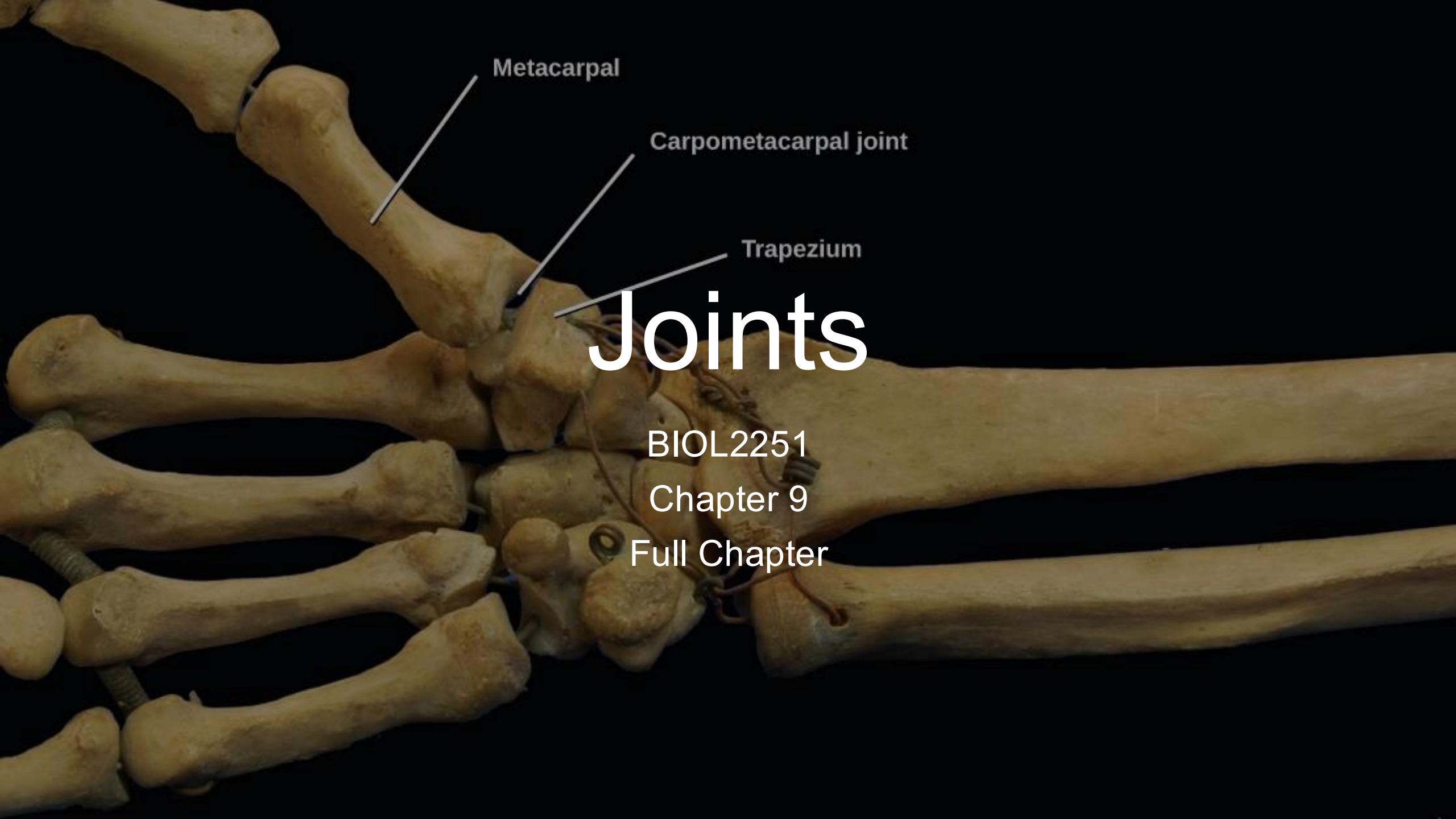


An anatomical dissection of a human hand joint, showing the metacarpals, carpal bones, and the trapezium. The bones are light-colored and set against a dark background. Labels with leader lines point to specific structures: 'Metacarpal' points to a long bone, 'Carpometacarpal joint' points to the joint between a carpal and metacarpal, and 'Trapezium' points to a small carpal bone.

Metacarpal

Carpometacarpal joint

Trapezium

Joints

BIOL2251

Chapter 9

Full Chapter

Mobility vs. Stability

Mobility vs. Stability

- **Instructions**

- Joints exist along a spectrum
 - Maximum mobility → Maximum stability
- **I will provide descriptions of joints, and you will direct me where to put each joint on this spectrum**
 - Justify your placement
- We will check and discuss at the end

- **Bloom's taxonomy = Remember, Understand**

Mobility vs. Stability

- Skull sutures → far left (maximum stability, zero mobility; fused bone edges protect the brain)
- Hip → left-center (very stable due to deep socket and strong ligaments, moderate mobility)
- Knee → center-left (stable but more mobile than hip; hinge motion primarily)
- Intervertebral joints → center (slight movement individually, but stability is essential for spinal protection)
- Elbow → center-left (stable hinge, reinforced by three bones locking together)
- Shoulder → far right (maximum mobility, sacrificed stability; shallow glenoid fossa)

Classify and Justify



Classify and Justify

- **Instructions**

- I have distributed individual worksheets
- On the front page, there is a table for you to fill
- **I will provide a list of 10 joint examples**
- Classify joints based on:
 - **Structural type (fibrous, cartilaginous, synovial)**
 - **Subtype (suture, gomphosis, syndesmosis, synostosis, synchondrosis, symphysis)**
 - **Functional type (synarthrosis, amphiarthrosis, diarthrosis)**
- Compare work with a neighbor, then we will complete the table on the board

- **Bloom's taxonomy = Remember, Understand**

Classify and Justify

- Joint between cranial bones
- Tooth in socket
- Distal tibiofibular joint
- Epiphyseal plate (growing)
- Joint between the pubis of the
os coxae
- Skull suture in mature adult
- Manubriosternal joint
- Intervertebral discs
- Hip joint
- Carpal-metacarpal joint

Classify and Justify

Joint	Structural Type	Subtype	Functional Type
Joint between cranial bones	Fibrous	Suture	Synarthrosis
Tooth in socket	Fibrous	Gomphosis	Synarthrosis
Distal tibiofibular joint	Fibrous	Syndesmosis	Amphiarthrosis
Epiphyseal plate (growing)	Cartilaginous	Synchondrosis	Synarthrosis
Joint between the pubis of the ox coxae	Cartilaginous	Symphysis	Amphiarthrosis
Fusion between two bones	Fibrous	Synostosis	Synarthrosis
Manubriosternal joint	Cartilaginous	Synchondrosis	Synarthrosis
Intervertebral discs	Cartilaginous	Symphysis	Amphiarthrosis
Hip joint	Synovial		Diarthrosis
Carpal-metacarpal	Synovial		Diarthrosis

Synchondrosis = hyaline cartilage, typically immovable; symphysis = fibrocartilage, slightly movable

Syndesmosis = fibrous but bones connected by ligament/interosseous membrane, slightly movable

Synostosis = what a suture becomes when it fully ossifies — same location, different classification across a lifetime

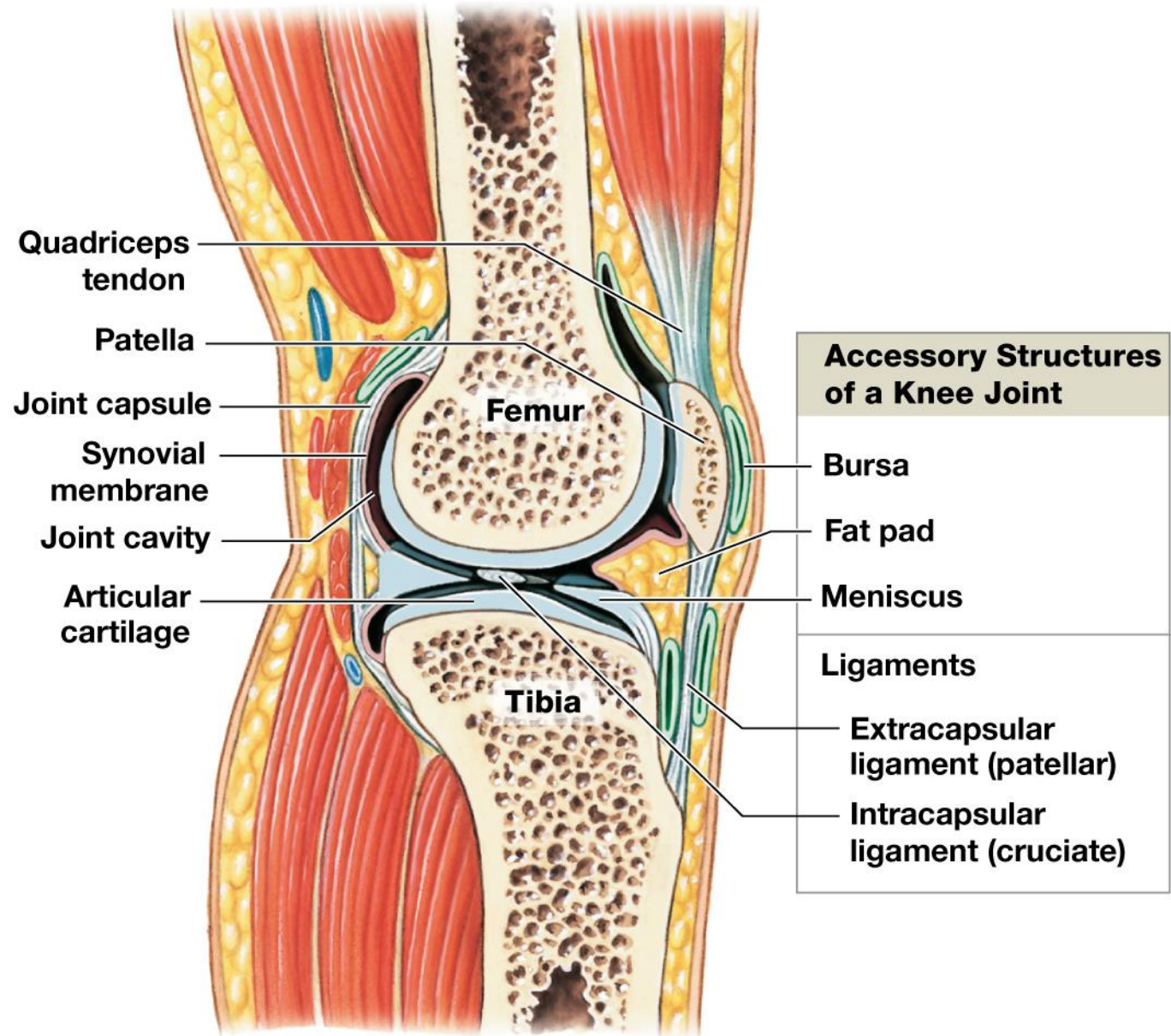


Build-a-Synovial Joint

Build-a-Synovial Joint

- **Instructions**

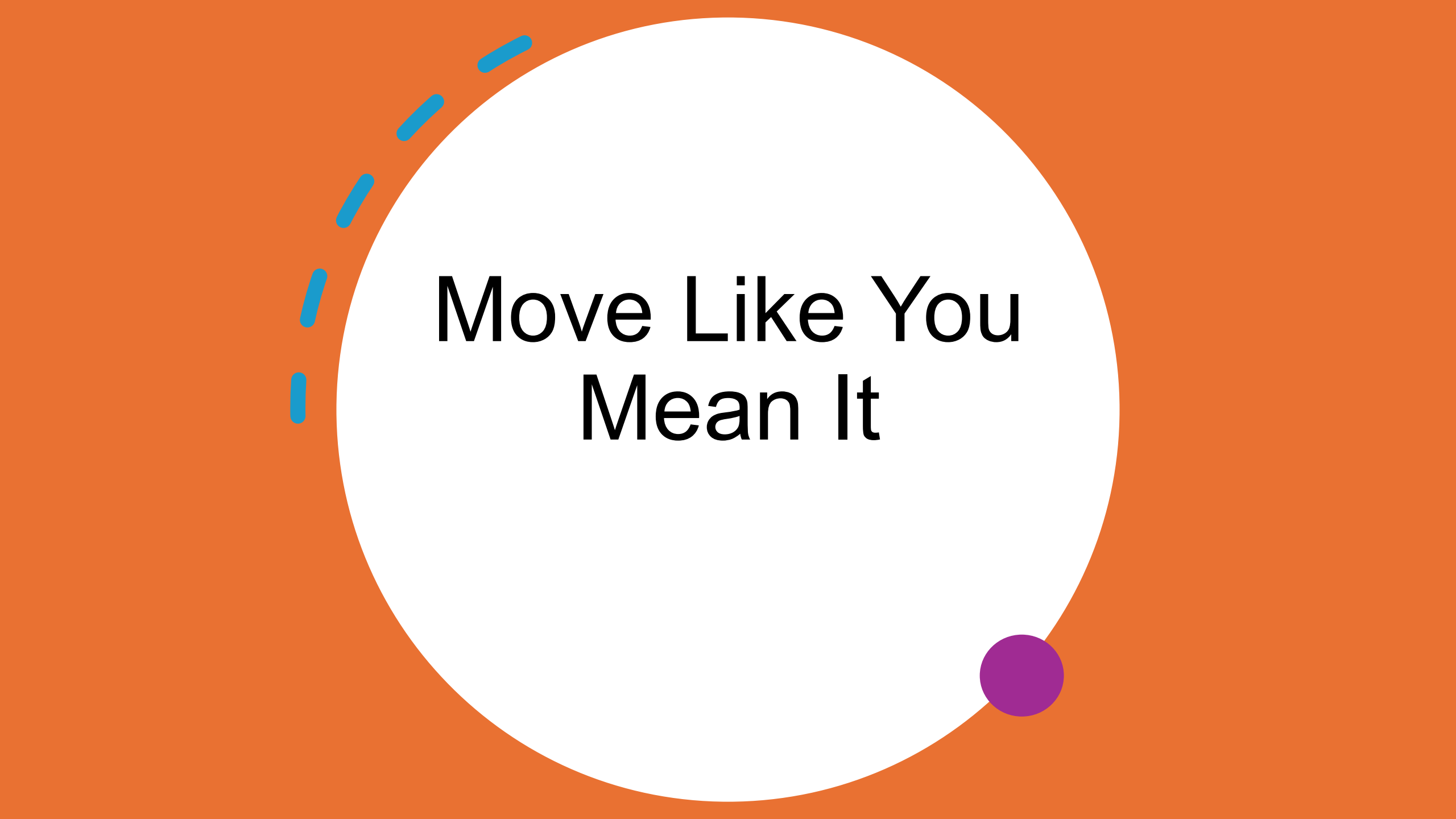
- I have distributed individual worksheets
- On the back page, there is a **synovial joint diagram for you to label**
- Complete the diagram on individual sheets, but you may work in pairs or groups
 - **Write a one-sentence function for each structure**
- Check your work with neighboring pairs or neighbors
- Debrief as a class
- **Bloom's taxonomy = Understand, Apply**



b Knee joint, sagittal section

Build-a-Synovial Joint

Structure	Function
Articular cartilage	Covers bone ends; reduces friction and absorbs compressive shock
Joint capsule (fibrous layer)	Encloses the joint; provides structural integrity and stability
Synovial membrane	Lines the inner capsule; secretes synovial fluid
Joint cavity	Space filled with synovial fluid; allows movement
Ligaments (extracapsular)	Reinforce the joint on the outside of the capsule (e.g., collateral ligaments of the knee)
Ligaments (intracapsular)	Reinforce the joint from within the capsule (e.g., ACL/PCL of the knee); covered by synovial membrane
Bursa	Fluid-filled sac that reduces friction between tendons/skin and bone near the joint
Fat pad	Cushions and fills irregular spaces within the joint capsule



Move Like You
Mean It

Move Like You Mean It

- **Instructions**

- I will call out a movement term
 - **You perform that movement using your own body**
- I **will then call out its paired opposite**
 - **Perform that opposing movement using your own body**
- We will review each term and movement at the end
- **Bloom's taxonomy = Understand, Apply**

Movement Term	Prompt	Description
Flexion	"Bend your elbow: what happened to the joint angle?"	Decreases joint angle; occurs at hinge joints (elbow, knee).
Extension	"Straighten your arm again."	Increases joint angle; opposite of flexion.
Hyperextension	"Look up toward the ceiling."	Extending beyond anatomical position; neck and back can hyperextend.
Abduction	"Raise your arm sideways away from your body."	Moves limb away from midline; shoulder and hip joints.
Adduction	"Bring your arm back to your side."	Moves limb toward midline.
Medial rotation	"Turn your leg so your toes point inward."	Rotation toward the midline.
Lateral rotation	"Now turn them outward."	Rotation away from midline.
Pronation / Supination	"Rotate your forearm palm down (pronation), then palm up (supination)."	Radius crosses over ulna in pronation; parallel in supination.
Dorsiflexion / Plantar flexion	"Flex and point your foot."	Dorsiflexion = up toward shin; plantar flexion = down like pressing a gas pedal.
Inversion / Eversion	"Roll your foot inward/outward."	Inversion turns sole medially; eversion laterally.
Elevation / Depression	"Shrug shoulders up and down."	Elevation lifts; depression lowers.
Protraction / Retraction	"Push your jaw forward, then pull it back."	Protraction = forward movement; retraction = backward.
Opposition / Reposition	"Touch thumb to pinky, then release."	Opposition enables grasp; reposition returns to neutral.

Synovial Joint Speed Match



Synovial Joint Speed Match

- **Instructions**

- **Pairs will receive a set of joint type names, movement descriptions, and anatomical examples**
 - **Correctly match all three columns**
 - First three pairs to submit the correct matches earn a prize
 - Debrief as a class
- **Bloom's taxonomy = Remember, Apply**

Synovial Joint Speed Match

Joint Type	Movement Allowed	Example
Gliding (plane)	Movement in multiple directions; minimal rotation	Intercarpal joints, intertarsal joints, acromioclavicular joint, sacro-iliac joint
Hinge	Flexion/extension only (uniaxial)	Elbow (humeroulnar), knee (modified), interphalangeal joints
Pivot	Rotation only around a longitudinal axis (uniaxial)	Atlantoaxial joint (C1–C2), proximal radioulnar joint
Condylar (ellipsoid)	Flexion/extension + abduction/adduction; biaxial	Radiocarpal (wrist), metacarpophalangeal joints
Saddle	Same as condylar but greater range; biaxial	Carpometacarpal joint of the thumb
Ball-and-socket	Multiaxial; all movements including rotation	Shoulder (glenohumeral), hip (coxal)



Joint Stability Showdown

Joint Stability Showdown

- **Instructions**

- Work in small groups (4-6 people) with whiteboards
- You will be assigned one of two comparisons
 - **Elbow vs. knee**
 - **Shoulder vs. hip**
- **Write at least four structural features for your joint**
 - **Elbow vs. knee: Joint type, bone fit, ligaments, menisci, overall stability**
 - **Shoulder vs. hip: socket depth, labrum, ligaments, muscle reinforcement, range of motion, dislocation risk**
- One group per pair share their work
- **Bloom's taxonomy = Analyze, Evaluate**

Joint Stability Showdown: Elbow vs. Knee

Feature	Elbow	Knee
Joint type	Hinge (humeroulnar) + pivot (humero radial + proximal radioulnar)	Modified hinge
Bone fit	Trochlea fits snugly into trochlear notch of ulna — bony interlocking provides stability	Femoral condyles rest on relatively flat tibial plateau — poor bone-to-bone fit
Ligaments	Radial and ulnar collateral ligaments reinforce sides	ACL, PCL (intracapsular), medial and lateral collateral ligaments, patellar ligament
Menisci	None	Medial and lateral menisci deepen the joint and absorb shock
Overall stability	Very stable due to bony interlocking	Less stable than elbow; relies heavily on ligaments and muscle tone

Joint Stability Showdown: Shoulder vs. Hip

Feature	Shoulder (Glenohumeral)	Hip (Coxal)
Socket depth	Shallow glenoid fossa; only ~1/3 of humeral head is covered	Deep acetabulum covers most of femoral head
Labrum	Glenoid labrum slightly deepens socket	Acetabular labrum significantly deepens the socket
Ligaments	Relatively weak glenohumeral ligaments	Very strong ligaments (iliofemoral, pubofemoral, and ischiofemoral)
Muscle reinforcement	Rotator cuff muscles are primary stabilizers	Massive muscles provide strong reinforcement
Range of motion	Greatest of any joint — multiaxial and very wide	Multiaxial but more restricted than shoulder
Dislocation risk	High — commonly-dislocated joint	Low — very stable, rare dislocation

Disc Dilemma



Disc Dilemma

- **Instructions**

- I will present a brief scenario
 - *A 45-year-old patient comes in reporting sharp radiating pain down their right leg. Imaging shows a problem with the disc between L4 and L5. The patient says, 'My doctor told me I have a herniated disc, but my chiropractor said it's a vertebral subluxation — which is it?'*
 - Think about the anatomy of the issue
 - **What is the patient experiencing?**
- **Bloom's taxonomy = Understand, Analyze**

Disc Dilemma

Intervertebral joint structure:

- The joint between vertebral bodies is a **symphysis** — a cartilaginous joint with a fibrocartilage pad called the **intervertebral disc**
- Each disc has two components:
 - **Nucleus pulposus** — the gel-like inner core, provides shock absorption; remnant of the embryonic notochord
 - **Annulus fibrosus** — the tough outer fibrocartilage ring, holds the nucleus in place and binds the vertebrae together

Disc Dilemma

Slipped disc vs. herniated disc:

- "Slipped disc" is a **misnomer** — the disc does not actually slip out of place. The entire disc never moves.
- A **herniated disc** occurs when the nucleus pulposus pushes through a tear or weakness in the annulus fibrosus, allowing the gel to bulge or extrude outward
- This herniated material can compress adjacent spinal nerve roots, causing radiating pain, numbness, or weakness in the distribution of that nerve (in L4–L5, pain commonly radiates down the leg as in sciatica)
- The correct term is **herniated disc**; the patient and their physical therapist should be corrected

Disc Dilemma

- **Vertebral subluxation complex** is a chiropractic term for a misaligned or dislocated vertebra
 - Definition from the WHO: A lesion or dysfunction in a joint or motion segment in which alignment, movement integrity and/or physiological function are altered, although contact between joint surfaces remains intact. It is essentially a functional entity, which may influence biomechanical and neural integrity.
 - In clinical practice, chiropractic "vertebral subluxation" is a diagnosis looking for a disease
 - While spinal manipulation may offer temporary relief for mechanical back pain (similar to physical therapy), the claim that it treats internal ailments is clinically unsubstantiated

Arthritis Clinic

Arthritis Clinic

- **Instructions**

- Work in small groups (4-6 people)
- I will distribute sheets with three patient descriptions and 12 statements to sort
 - **Classify each statement to the applicable patient and identify the type of arthritis each patient likely has**
- 3-4 groups will share their reasoning
- **Explain not only why you selected your condition, but why you ruled out the other two conditions**
- **Bloom's taxonomy = Understand, Apply, Analyze**

Arthritis Clinic

- **Patient 1:** "Mathilda, age 71, reports deep aching pain in her right hip that has worsened over the years. She says the stiffness she feels in the morning is gone within about 20 minutes of getting up and moving around."
- **Patient 2:** "Jemaine, age 44, woke up with intense pain and swelling in his right hallux after a weekend of rich food and alcohol. His doctor ordered blood work and found elevated levels of a metabolic waste product."
- **Patient 3:** "Moirra, age 40, has had pain and swelling in both hands and both wrists for nearly a year. Her morning stiffness lasts well over an hour, and she has been feeling unusually fatigued."

Arthritis Clinic

Patient 1 — Osteoarthritis	Patient 2 — Gouty Arthritis	Patient 3 — Rheumatoid Arthritis
Cartilage gradually wears down over time	Crystals deposit in the synovial fluid	The immune system attacks the joint's own tissues
Most common in older adults and weight-bearing joints	A metabolic disorder underlies this condition	Joint involvement tends to be symmetrical
Symptoms are usually limited to the affected joint	Uric acid accumulates in the blood	Prolonged morning stiffness lasting an hour or more
Morning stiffness that fades quickly after moving	Often affects the big toe	The synovial membrane is the primary target

Arthritis Clinic

- **Mathilda (OA):** Gradual onset in a weight-bearing joint in an older adult is the central clue. The brief morning stiffness is a classic distinguishing feature from rheumatoid arthritis, where stiffness persists much longer. No systemic symptoms and no mention of metabolic issues rule out the other two.
- **Jemaine (Gout):** Sudden acute onset, the specific joint (big toe), the dietary trigger, and the elevated metabolic waste product (uric acid) collectively point to gout. Students should recognize "metabolic waste product" as uric acid even without it being named. His age and the episodic nature also help distinguish it from OA.
- **Moira (RA):** Bilateral, symmetrical involvement in a relatively young patient, prolonged morning stiffness, and systemic fatigue are the distinguishing hallmarks. The hands and wrists are classic sites. The autoimmune and synovial membrane statements belong here because RA is driven by the immune system attacking the synovium, not by wear or metabolic dysfunction.